

Haoyi Zhu

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EDUCATION

University of Science and Technology of China (USTC), China <i>Ph.D.</i> in Computer Science (Advisor: Prof. Xiaogang Wang)	Sep. 2023 – Present
Shanghai Jiao Tong University (SJTU), China <i>B.E.</i> in Artificial Intelligence Honor Class (Advisor: Prof. Cewu Lu)	Sep. 2019 – Jun. 2023

SELECTED AWARDS

China3DV Academic Rising Star Award Candidate	Apr. 2026
Outstanding Paper Award, ICCV 2025 RIWM Workshop	Oct. 2025
National Scholarship	Oct. 2025
First Price Academic Scholarship, USTC	Sep. 2025
XingQi Intern, Shanghai AI Lab (6/160+)	Apr. 2025
Outstanding Paper Award, NeurIPS 2022	Dec. 2022

INTERNSHIPS

Efficient AI Lab, NVIDIA <i>Research Intern</i> (Advisor: Dr. Enze Xie & Prof. Song Han) World Model	Dec. 2025 - Present
Shanghai AI Lab <i>Research Intern</i> (Advisor: Prof. Tong He & Prof. Wanli Ouyang) - World Model - Spatial Intelligence	Sep. 2023 - Nov. 2025
MVIG, Shanghai Jiao Tong University (SJTU) <i>Research Intern</i> (Advisor: Dr. Hao-Shu Fang & Prof. Cewu Lu) - Human Pose Estimation - General Robot Manipulation	Sep. 2019 - Jun. 2023
GEAR Lab, NVIDIA <i>Research Intern</i> (Advisor: Dr. Guanzhi Wang & Dr. Jim Fan) General Open-Ended Embodied Agent	Feb. 2022 - Feb. 2023
Xu Lab, Carnegie Mellon University (CMU) <i>Research Intern</i> (Advisor: Prof. Min Xu) AI for Biology	Apr. 2021 – Feb. 2022

PUBLICATIONS

Total Citations: 2726

Journal Papers

PonderV2: Pave the Way for 3D Foundation Model with A Universal Pre-training Paradigm [[paper](#)] [[code](#)]
Haoyi Zhu*, Honghui Yang*, Xiaoyang Wu*, Di Huang*, Sha Zhang, Xianglong He, Tong He, Hengshuang Zhao, Chunhua Shen, Yu Qiao, Wanli Ouyang
IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), 2025

AlphaTracker: A Multi-Animal Tracking and Behavioral Analysis Tool [[paper](#)] [[code](#)]
Zexin Chen, Ruihan Zhang, Hao-Shu Fang, Yu E Zhang, Aneesh Bal, Haowen Zhou, Rachel R Rock, Nancy Padilla-Coreano, Laurel R Keyes, **Haoyi Zhu**, Yong-Lu Li, Takaki Komiyama, Kay M Tye, Cewu Lu
Frontiers in Behavioral Neuroscience, 2023

AlphaPose: Whole-Body Regional Multi-Person Pose Estimation and Tracking in Real-Time [[paper](#)] [[code](#)]
Hao-Shu Fang, Jiefeng Li, Hongyang Tang, Chao Xu*, **Haoyi Zhu***, Yuliang Xiu, Yong-Lu Li, Cewu Lu
[>8.3K GitHub Stars] *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 2022

Conference Papers (Selected)

OmniWorld: A Multi-Domain and Multi-Modal Dataset for 4D World Modeling [paper] [website] [code]

..., **Haoyi Zhu**, ... (19 authors)

arXiv preprint, 2025

WinT3R: Window-Based Streaming Reconstruction with Camera Token Pool [paper] [website] [code]

Yifan Wang, Jianjun Zhou, **Haoyi Zhu**, Wenzheng Chang, Yang Zhou, Zizun Li, Junyi Chen, Jiangmiao Pang, Chunhua Shen, Tong He

arXiv preprint, 2025

π^3 : **Scalable Permutation-Equivariant Visual Geometry Learning** [paper] [website] [code]

Yifan Wang, Jianjun Zhou, **Haoyi Zhu**, Wenzheng Chang, Yang Zhou, Zizun Li, Junyi Chen, Jiangmiao Pang, Chunhua Shen, Tong He

arXiv preprint, 2025

DeepVerse: 4D Autoregressive Video Generation as a World Model [paper] [website] [code]

Junyi Chen, **Haoyi Zhu**, Xianglong He, Yifan Wang, Jianjun Zhou, Wenzheng Chang, Yang Zhou, Zizun Li, Zhoujie Fu, Jiangmiao Pang, Tong He

arXiv preprint, 2025

VQ-VLA: Improving Vision-Language-Action Models via Scaling Vector-Quantized Action Tokenizers [paper] [website] [code]

Yating Wang, **Haoyi Zhu**, Mingyu Liu, Jiange Yang, Hao-Shu Fang, Tong He

International Conference on Computer Vision (ICCV), 2025

Aether: Geometric-Aware Unified World Modeling [paper] [website] [code]

Haoyi Zhu, Yifan Wang, Jianjun Zhou, Wenzheng Chang, Yang Zhou, Zizun Li, Junyi Chen, Chunhua Shen, Jiangmiao Pang, Tong He

International Conference on Computer Vision (ICCV), 2025

[Outstanding Paper Award & Oral Presentation] ICCV 2025 Workshop on Reliable and Interactive World Models (RIWM)

CoMo: Learning Continuous Latent Motion from Internet Videos for Scalable Robot Learning [paper]

Jiange Yang, Yansong Shi, **Haoyi Zhu**, Mingyu Liu, Kaijing Ma, Yating Wang, Gangshan Wu, Tong He, Liming Wang

arXiv preprint, 2025

SPA: 3D Spatial-Awareness Enables Effective Embodied Representation [paper] [website] [code]

Haoyi Zhu, Honghui Yang, Yating Wang, Jiange Yang, Limin Wang, Tong He

International Conference on Learning Representations (ICLR), 2025

Tra-MoE: Learning Trajectory Prediction Model from Multiple Domains for Adaptive Policy Conditioning [paper] [code]

Jiange Yang, **Haoyi Zhu**, Yating Wang, Gangshan Wu, Tong He, Limin Wang

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025

Point Cloud Matters: Rethinking the Impact of Different Observation Spaces on Robot Learning [paper] [website] [code]

Haoyi Zhu, Yating Wang, Di Huang, Weicai Ye, Wanli Ouyang, Tong He

Advances in Neural Information Processing Systems (NeurIPS), 2024

RH20T: A Comprehensive Robotic Dataset for Learning Diverse Skills in One-Shot [paper] [website] [code]

Hao-Shu Fang, Hongjie Fang, Zhenyu Tang, Jirong Liu, Chenxi Wang, Junbo Wang, **Haoyi Zhu**, Cewu Lu

IEEE International Conference on Robotics and Automation (ICRA), 2024

UniPAD: A Universal Pre-Training Paradigm for Autonomous Driving [paper] [code]

Honghui Yang, Sha Zhang, Di Huang, Xiaoyang Wu, **Haoyi Zhu**, Tong He, Shixiang Tang, Hengshuang Zhao, Qibo Qiu, Binbin Lin, Xiaofei He, Wanli Ouyang

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024

X-NeRF: Explicit Neural Radiance Field for Multi-Scene 360° Insufficient RGB-D Views [paper] [code]

Haoyi Zhu, Hao-Shu Fang, Cewu Lu

IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2023

MineDojo: Building Open-Ended Embodied Agents with Internet-Scale Knowledge [paper] [website] [code]

Linxi Fan, Guanzhi Wang, Yunfan Jiang, Ajay Mandlekar, Yuncong Yang, **Haoyi Zhu**, Andrew Tang, De-An Huang, Yuke Zhu, Anima Anandkumar

[Outstanding Paper Award] *Advances in Neural Information Processing Systems (NeurIPS)*, 2022

For the full publication list, please visit my [Google Scholar](#) page.

SELECTED PROJECTS

Sana World Model

Minute-Scale Video World Model with Linear-Complexity Attention

Dec. 2025 – Present

- Developed an efficient world model that extends Sana Video from 5-second clips to **minute-scale (961 frames), 720p, camera-controllable** video generation with only 1.6B parameters. Replaced the Wan VAE with an LTX2 VAE (compression stride $8 \times 32 \times 32$), achieving **$3.5 \times$ fewer tokens** at 720p than prior work at 480p.
- Proposed **Frame-wise Gated DeltaNet (GDN)**, adapting token-level GDN to process entire spatial frames simultaneously. Derived a zero-overhead RMSNorm stabilization trick that bounds the spectral radius of the transition matrix, resolving catastrophic NaN explosions in BF16 training. Designed bidirectional and chunk-causal GDN variants for training and autoregressive inference, respectively.
- Introduced a **hybrid GDN-Softmax attention** architecture (15 GDN + 5 Softmax layers), and a **dual-rate camera conditioning** mechanism combining UCPE relative ray encoding (per latent frame) with Plücker coordinate injection (per raw frame), enabling precise 6-DoF trajectory control within the linear-attention backbone.
- Designed a four-stage progressive training pipeline (VAE finetuning $\rightarrow$ GDN distillation $\rightarrow$ length extension $\rightarrow$ camera control) with careful initialization for linear-attention equivalence, enabling stable knowledge transfer from pretrained checkpoints.

Aether

Scalable and Generalizable Foundation World Model

Nov. 2024 – Nov. 2025

- Proposed **Aether**, a geometry-aware unified world model. **Aether** features 3 core capabilities: (1) 4D dynamic reconstruction, (2) action-conditioned video prediction, and (3) goal-conditioned visual planning. Building upon video generation models, our framework demonstrates unprecedented synthetic-to-real generalization despite *never* observing real-world data during training. Won the **Outstanding Paper Award** at ICCV 2025 RIWM workshop.
- Developed **DeepVerse**, a 4D auto-regressive world model. **DeepVerse** can fantasize the entire world behind images and enable free exploration through interaction. It demonstrates exceptional capabilities in modeling dynamics, geometry, and physical laws.
- Introduced π^3 , a novel feed-forward neural network that revolutionizes visual geometry reconstruction by eliminating the need for a fixed reference view. With a fully permutation-equivariant architecture, π^3 achieves SOTA performance on a wide range of tasks, including camera pose estimation, monocular/video depth estimation, and dense point map estimation.
- Mentored several undergraduate and junior Ph.D. students on robot learning topics, including action tokenizers, cross-embodiment policies, point cloud policies, *etc.*

MineDojo

Building Open-Ended Embodied Agents with Internet-Scale Knowledge

Feb. 2022 – Jul. 2022

- Contributed to the **MineDojo** project, a new framework designed to build generally capable agents with internet-scale knowledge in Minecraft. Won the **Outstanding Paper Award** at NeurIPS 2022.
- Co-developed the YouTube database, which comprises over 730,000 narrated Minecraft videos, totaling approximately 300,000 hours of content and 2.2 billion English transcripts.
- Constructed MineCLIP, a video CLIP foundation model that acts as a learned reward function, enabling agents to solve diverse open-ended tasks specified in natural language.
- Authored APIs for three databases (YouTube, Minecraft Wiki, and Reddit), along with the corresponding documentation and part of the official project website.

ACADEMIC SERVICES

Reviewer for journals, including IJCV, TMLR, *etc.*

Reviewer for conferences, including ICCV, CVPR, NeurIPS, ICLR, WACV, ICIP, *etc.*

INVITED TALKS

- 2025/08/26, [Prof. Lin Shao](#)'s Lab, NUS, "World Model and Spatial Intelligence"
- 2025/06/12, 3D CV Workshop (3D视觉工坊), "Towards Generalizable and Scalable Spatial Embodied Intelligence" [[link](#)]
- 2025/05/22, Embodied Intelligent Mind (具身智能之心), "Towards Generalizable and Scalable Spatial Embodied Intelligence"
- 2025/04/22, [Prof. Yanyong Zhang](#)'s Lab, USTC, "Towards Scalable World Model for General Embodied AI"
- 2025/03/27, Graduate Academic Forum, USTC, "Towards Spatial Intelligence: From 3D Vision to Embodied AI"
- 2024/12/27, ZhiXingXing Embodied AI Frontier Lecture (智猩猩具身智能前沿讲座), "Towards Spatial Intelligence: From 3D Vision to Embodied AI" [[link](#)]